Behav. Res. Ther. Vol. 20, pp. 391 to 396, 1982
Printed in Great Britain.

0005-7967/82/040391-06$03.00/0

SECOND-ORDER 'FEAR' CONDITIONING IN HUMANS: PERSISTENCE OF CR_2 FOLLOWING EXTINCTION OF CR_1

GRAHAM C. L. DAVEY* and TABITHA ARULAMPALAM
Department of Social Science and Humanities, The City University, Northampton Square, London EC1V 0HB, U.K.

(*Received* 30 *September* 1981)

Summary—Two groups of subjects were given pairings of a slide of a triangle (CS_1) with an aversive loud noise (US). After first-order conditioning a different figure (CS_2^+) was paired with CS_1, whilst a third figure acted as a non-paired control (CS_2^0). Conditioned second-order skin conductance responses (SCRs) were established to CS_2^+ for all subjects who had undergone first-order conditioning. Following second-order conditioning, SCRs to CS_1 were extinguished and CS_2^+ was subsequently tested for retention of its SCR evoking properties. For the group of subjects who believed they had an avoidance response available during the second-order conditioning phase, CR_2 failed to survive extinction of CR_1. However, for the group of subjects who had no perceived control over US presentation, extinction of CR_1 did not eliminate CR_2. This latter finding is consistent with associative analyses of second-order conditioning in animals and has clinical implications for the treatment of fear-related problems.

INTRODUCTION

Recent accounts of human classical conditioning have deviated from traditional associative views by postulating the involvement of cognitive factors which mediate conditioning (e.g. Brewer, 1974; Dawson and Furedy, 1977). These accounts have tended to ignore the role played by the associative substructure underlying learning in favour of mediating variables which stress the importance of such states as 'awareness', 'compliance', 'expectancy' etc. (cf. Brewer, 1974), and the movement of human classical conditioning theory in this direction has tended to take place in isolation from current developments in animal associative learning. It is our view that a study of human classical conditioning in terms of associative factors has been sorely neglected—especially since animal studies have recently refined experimental techniques which allow a greater understanding of the associations formed during conditioning (Davey, 1981, pp. 215–242; Dickinson, 1980; Rescorla, 1980).

For instance, in a series of experiments, Rescorla and his colleagues have been able to demonstrate the types of associations formed during first- and second-order conditioning in rats (cf. Rescorla, 1980). For example, if a tone (conditioned stimulus, CS) is paired with electric shock (unconditioned stimulus, US), subsequent presentations of the tone will suppress concurrent bar-pressing for food (the conditioned emotional response, CER). However, if the subject is then given off-the-baseline habituation training to the US, re-exposure to the CS fails to produce a CR (Rescorla, 1973). This result is evidence for the formation of CS–US associations during conditioning, since the CR appears to be mediated through a representation of the US, and these findings with first-order conditioning have since been substantiated in appetitive conditioning situations (Holland and Rescorla, 1975a; Holland and Straub, 1979). However, when second-order conditioning is established through pairing of CS_2 with CS_1 (Pavlov, 1927) different associations appear to mediate CR_2. When responding to a second-order CS_2 has been established through its pairing with an already conditioned CS_1, the following post-conditioning manipulations do not eliminate CR_2 during subsequent CS_2-alone test presentations: (i) habituation to the original US in aversive conditioning (Rescorla, 1973); (ii) conditioned aversion to the US in appetitive conditioning (Holland and Rescorla, 1975a); and (iii)

* To whom all correspondence should be addressed.

extinction of the response to CS_1 (Holland and Rescorla, 1975b; Rizley and Rescorla, 1972). These results suggest that during second-order conditioning, subjects tend to associate CS_2 with their response to CS_1 (an S–R association), hence representations of CS_1 and the US play no part in generating CR_2 (Rescorla, 1977, 1980).

The present study is designed to investigate the associations formed during second-order electrodermal conditioning in humans using one of the techniques just described. Subjects were given pairings of a slide of a geometric shape (CS_1) with an aversive loud noise (US). Following this, a second slide (CS_2^+) was paired with CS_1. Subsequently, the response to CS_1 was extinguished and subjects were finally tested for the survival of CR_2^+ following this manipulation.

METHOD

Subjects

The *S*s were 27 undergraduate volunteers of both sexes, whose ages ranged from 19 to 23 yr. They were all naive as to the purpose of the experiment and a majority were *S*s with no knowledge of psychology. Subjects were divided randomly into three groups of nine. These groups were labelled Group E (experimental), Group A (avoidance) and Group C (control).

Apparatus

Skin conductance (SCR) was measured by bipolar placement of anoidized thimble electrodes attached to the distal phalanx of the middle finger and ring finger of the left hand. These electrodes projected changes in SCR onto a Washington 400 MD 2C polygraph via a constant voltage circuit. Slides used as stimuli were projected onto a screen approx. 1 m in front of the *S* and the size of the resulting picture was 30 × 60 cm. All *S*s wore earphones which emitted ambient white noises at a level of 65 ± 2dB. These earphones could also present 0.5-sec bursts of a 115dB 1000 Hz tone which was used as the US. For groups A and C a small telegraph key was situated on a table directly to the left of the *S*. Presentation of slides and auditory stimuli was controlled by solid-state logic programming equipment located on a bus–bar rack system.

Procedure

On introduction to the experimental room, *S*s were given written instructions which told them that the experiment was designed to measure sweat-gland activity. They were seated in a comforable chair facing the screen on which slides of various outline figures were to be displayed. Subjects were then fitted with the thimble electrodes and earphones, and were told to expect brief written instructions that would be placed in front of them during the experiment. All *S* were initially told to relax and watch the figures projected onto the screen. Prior to the experimental conditions being instigated, all *S*s were given, in random order, three presentations of the US and three presentations of a slide showning an equilateral triangle. The latter lasted for 6 sec and was subsequently to be used as CS_1 for all *S*s. The experiment itself consisted of four phases.

Phase 1. This phase consisted of first-order electrodermal conditioning in which, for Groups E and A, the 115dB tone always followed the 6-sec presentation of the slide showing a triangle (CS_1). They received 10 pairings of CS_1 and US, with each trial being separated by a 30-sec inter-trial interval. Group C also received 10 presentations of CS_1 and tone but these were presented on independent variable time 30-sec schedules (a random control procedure). All *S*s were pre-warned about the contingencies they were to receive through written instructions placed in front of them.

Phase 2. This phase consisted of second-order conditioning in which the slide of the triangle (CS_1) always immediately followed the 6-sec presentation of a slide of an outline drawing of a telephone (CS_2^+). Randomly interspersed amongst these pairings was the individual presentation of an outlined drawing of a kitchen tap (CS_2^0). This was a second-order control stimulus which was never paired with CS_1. Approximately half of the *S*s had the significance of the second-order stimuli reversed such that the kitchen tap was CS_2^+ and the telephone was CS_2^0. This phase lasted for five presentations of CS_2^+ paired with CS_1, and five independent presentations of CS_2^0. There was a 30-sec inter-trial interval between CS_2^+–CS_1 pairings and CS_2^0 presentations. All Ss were informed of the relationship between the two new stimuli and CS_1 prior to the start of this phase.

However, during this phase, Group A was also told that the tone (US) could be avoided if they pressed the telegraph key during the presentation of CS_1. Group C was simply told to press the key whenever they saw CS_1, and for Group E the key was removed and no response was available to them. The US was never presented to any group during this phase, but in effect Group A had pairings of CS_2^+ and CS_1 whilst believing they were avoiding the US by pressing the lever during CS_1. Groups E and C simply had pairings of CS_2^+ and CS_1.

Phase 3. In this phase all groups were given 10 trials in which CS_1 was presented alone. Subjects were informed of this fact prior to this phase, and in effect it was designed to extinguish conditioned SCRs to CS_1. There was a 30-sec inter-trial interval between presentations.

Phase 4. In the final phase all groups were given eight individual slide presentations, two of these presentations were CS_2^+ and two were of CS_2^0. The remainder were interspersed randomly amongst these, and were slides which were new to the *S* (outline drawings of an aeroplane, square, star and a circle). There was a 30-sec interval between each slide presentation.

Response measurement

Electrodermal responses were monitored during each slide presentation and for selected 6-sec periods during the inter-trial interval. In order to minimize the effects of drift in SCR levels and spontaneous responses resulting from the bias and polarization potentials of the electrodes, all SCRs occurring during slide presentations were compared with SCRs measured in a randomly selected period (6-sec) during proximal inter-trial intervals. In the first three phases of the experiment only SCR measurements taken from the last three presentations of each stimulus in each phase contributed to the data to be presented. Two inter-trial interval readings were also taken from this final period. For Phase 4, SCR measurements were taken from all stimulus presentations plus two inter-trial interval readings taken 15 sec after the termination of two randomly selected trials.

A response was considered as an upward curve of the SCR within 5-sec of the onset of a stimulus and the magnitude of the response was calculated by measuring the distance between the trough and apex of the curve; all response changes which did not result in an upward curve of the SCR were given zero value.

The vast majority of trials with responses possessed only one response peak; only a handful possessed two response peaks and when this occurred only the first response was measured.

Finally, all SCR values were treated with a correction factor used to normalize SCR values across Ss. The formula for this is $\phi = SCR/SCR_{max}$ (Lykken, 1972) and SCR_{max} was taken from the *S*'s largest response to the tone during the pre-experimental habituation period.

RESULTS

Figure 1 shows ϕSCR values for all three groups for Phases 1, 2 and 4 of the experiment. First-order conditioning is indicated in both Groups A ($t = 2.431$, $P < 0.05$) and E($t = 5.638$, $P < 0.01$) when SCR measures during CS_1 and ITI are compared. Group C shows no difference in SCR between CS_1 and ITI measures ($t = 1.27$, $P > 0.05$). During second-order conditioning both Groups A and E exhibit second-order conditioning, with SCR during CS_2^+ being significantly higher than that during CS_2^0 (Group A, $t = 2.563$, $P < 0.05$; Group E, $t = 3.432$, $P < 0.01$) and higher than that during ITI (Group A, $t = 3.04$, $P < 0.05$; Group E, $t = 3.588$, $P < 0.01$). Responding to CS_2^0 was no higher than that recorded during ITI (Group A, $t = 0.863$, $P > 0.05$; Group E, $t = 0.447$,

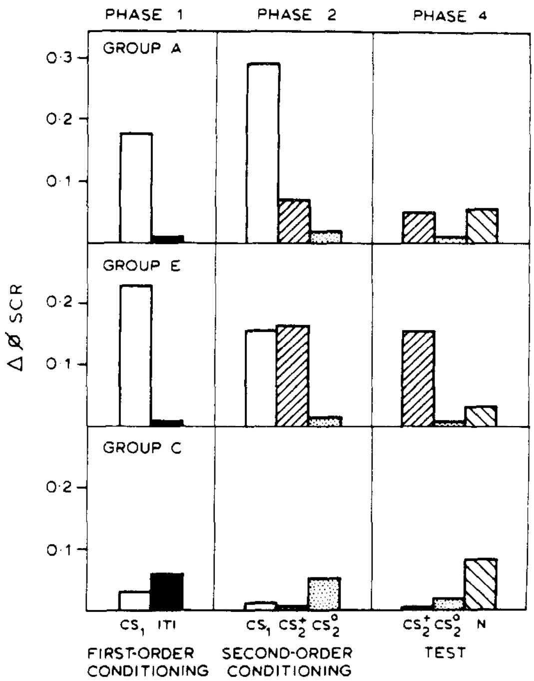

Fig. 1. Mean ϕSCR values for all three groups for Phases 1, 2 and 4 of the experiment. CS response values during Phases 1 and 2 are calculated from the last three trials in each phase. In Phase 4, means are calculated from the two presentations of CS_2^+ and CS_2^o and the four presentations of novel stimuli (N). ITI values are a mean of the two samples taken during each phase. [$\Delta\phi$SCR = change in SCR (range-corrected $\mu\Omega$)].

$P > 0.05$). Although Fig. 1 suggests that response to CS_1 is greater in Group A than Group E this difference did not reach statistical significance ($t = 1.564$, $P > 0.05$). Similarly, although the mean SCR to CS_2^+ was smaller for Group A than Group E, this difference just failed to reach statistical significance ($t = 1.8826$, $P > 0.05$). Since the original US was never presented during this phase, the SCR acquired by CS_2^+ must have resulted from the reinforcing properties of its pairing with CS_1. All *S*s in Group A reported in a post-experimental interview that they believed the key-pressing response avoided US presentation.

All groups ended Phase 3 by showing no differential responding to CS_1 when CS_1 values for the last three trials were compared with ITI values (Group A, $t = 1.643$, $P > 0.05$; Group E, $t = 1.703$, $P > 0.05$; Group C, $t = 1.223$, $P > 0.05$).

Phase 4 was used to test for a retention of conditioned responding to CS_2^+ even though the response to the original stimulus (CS_1) has been extinguished. During Phase 4, for Group A, SCR to CS_2^+ was not significantly different to that emitted to CS_2^o ($t = 1.71$, $P > 0.05$); nor to the novel stimuli ($t = 0.0167$, $P > 0.05$), nor during the ITI ($t = 0.873$, $P > 0.05$). This suggests that the SCR to CS_2^+ did not survive the extinction of responding to CS_1. However, in Group E, responding to CS_2^+ did appear to survive this manipulation. SCR to CS_2^+ was greater than that to CS_2^o ($t = 3.164$, $P < 0.05$) and the novel stimuli ($t = 2.69$, $P < 0.05$) and ITI responding ($t = 2.873$, $P < 0.05$).

DISCUSSION

The results of this experiment demonstrate (i) that second-order electrodermal conditioning can readily be acquired by human subjects; and (ii) under some conditions, this second-order response can survive subsequent extinction of the respone to the original first-order CS. However, the latter result was not found when subjects had an avoidance response available to them in Phase 2 of the experiment. The reasons why an avoidance response should preclude survival of CR_2^+ is not immediately clear. However, one possibility stems from the clinical literature. Seligman (1976) has pointed out that clinical evidence suggests that when a situation is controllable, anxiety responses tend to focus on to stimuli immediately preceding the aversive event. This observation tends to be borne out by the results of the present experiment. Subjects in Group A, who believed they had some control over the situation, generally exhibited a larger SCR to CS_1 in Phase 2 than did subjects in Group E (see Fig. 1). In fact, seven out of nine subjects in Group A showed an increase in SCR to CS_1 between Phases 1 and 2, whilst only two subjects from Group E did so. This focusing of the SCR response to stimuli immediately preceding the aversive event may leave the response to CS_2^+ particularly weak and unable to survive subsequent maniuplations. The increase in mean SCR to CS_1 in Group A could not have been due solely to the observational and motor requirements of key pressing because Group C, who had a similar requirement, did not show an increase in SCR to CS_1 in Phase 2.

Nevertheless, the second-order response did survive extinction of CR_1 for those subjects who did not have a supposed avoidance response available and this is a finding which is consistent with most animal studies of second-order conditioning (cf. Rescorla, 1980). The conclusion to be drawn from this is that CS_2–CS_1 associations do not play a major role in mediating the appearance of CR_2. The survival of CR_2 following extinction of the response to CS_1 implies that subjects had either formed an S–R association between CS_2 and their response to CS_1, or had associated CS_2 directly with a representation of the US. Further experimentation should clarify these possibilities. For instance, if subjects form CS_2–US associations, CR_2 should not survive habituation to the US (e.g. Holland and Rescorla, 1975a).

Finally, the finding that second-order 'fear' response can survive extinction of the first-order response has a number of implications for clinicians dealing with fear and anxiety-related problems. First, the fact that extinction of CR_1 does not necessarily result in the extinction of CR_2 may account for the relative persistence of some phobic and fear-related responses: reducing fear to one feature of a phobic event will not necessarily result in extinction of that response to other related features—especially when those other features have never been explicitly paired with an aversive US but gain their fear evoking properties through association with other conditioned aversive stimuli. Secondly, phobias are frequently characterized by situations that "the individual objectively recognizes should not be fear provoking" (Leitenberg, 1976, p. 124), yet the fear response is still evoked. If such fear responses are a result of second-order conditioning and are controlled by S–R associations (as they are in non-human animals) then substituting rational cognitions for irrational ones should have little effect on alleviating the fear response since the event that reinforced that association no longer influences or mediates the fear response. In fact, irrational cognitions may not necessarily act to maintain fear responses (Rimm *et al.*, 1977), but may simply be the individuals *post-hoc* attempt to 'explain' a reflexive fear reaction that has been established through higher-order conditioning.

REFERENCES

Brewer F. (1974) There is no convincing evidence for operant or classical conditioning in adult humans. In *Cognition and the Symbolic Processes* (Edited by Werner W. B. and Palermo D. S.). Halstead, New York.

Davey G. C. L. (1981) *Animal Learning and Conditioning*. Macmillan, London.

Dawson M. E. and Furedy J. J. (1976) The role of awareness in human differential autonomic classical conditioning: the necessary-gate hypothesis. *Psychophysiology* **13**, 50–53.

Dickinson A. (1980) *Contemporary Animal Learning Theory*. Cambridge Univ. Press, Massachusetts.

Holland P. C. and Rescorla R. A. (1975a) The effects of two ways of devaluing the unconditioned stimulus after first and second-order appetitive conditioning. *J. exp. Psychol.: Anim. Behav. Proc.* **1**, 355–363.

Holland P. C. and Rescorla R. A. (1975b) Second-order conditioning with food unconditioned stimulus. *J. comp. physiol. Psychol.* **88**, 459–467.

Holland P. C. and Straub J. J. (1979) Differential effects of two ways of devaluing the unconditioned stimulus after Pavlovian appetitive conditioning. *J. exp. Psychol.: Anim. Behav. Proc.* **5**, 65–78.

Leitenberg H. (Ed.) (1976) Behavioral approaches to treatment of neurosis. In *Handbook of Behavior Modification and Behavior Therapy*. Prentice-Hall, Englewood Cliffs, New Jersey.

LYKKEN D. T. (1972) Range correction applied to heart rate and GSR data. *Psychophysiology* **9**, 373–382.

PAVLOV I. P. (1927) *Conditioned Reflexes.* Oxford Univ. Press, Oxford.

RESCORLA R. A. (1973) Second-order conditioning: implication for theories of learning. In *Contemporary Approaches to Learning and Conditioning* (Edited by MCGUIGAN F. J. and LUMSDEN D.). Winston, New York.

RESCORLA R. A. (1977) Pavlovian second-order conditioning: some implications for instrumental behavior. In *Operant-Pavlovian Interactions* (Edited by DAVIS H. and HURWITZ H. M. B.). Erlbaum, Hillsdale, New Jersey.

RESCORLA R. A. (1980) *Pavlovian Second-Order Conditioning.* Erlbaum, Hillsdale, New Jersey.

RIMM D. C., JANDA L. H., LANCASTER D. W., NAHL M. and DITTMAR K. (1977) An exploratory investigation of the origin and maintenance of phobias. *Behav. Res. Ther.* **15**, 231–238.

RIZLEY R. C. and RESCORLA R. A. (1972) Associations in second-order conditioning and sensory preconditioning. *J. comp. physiol. Behav.* **81**, 1–11.

SELIGMAN M. E. P. (1976) *Helplessness.* Freeman, San Francisco.